

Prakhar Shekhar Parthasarthi

Software Engineer · Systems · Correctness · Applied AI infrastructure

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Shahjahanpur, UP · open to relocate

github.com/praxstack | linkedin.com/in/prakharshekhar | prax-portfolio-one.vercel.app

195k

GET rps · Java Redis

<0.3

ms p50

42

Redis tests

47%

p90 latency cut

SUMMARY

Software engineer focused on correctness and throughput. Amazon: p90 cut 47% (1.1s → 0.59s), conversion +33%, idempotent SQS at 43 bps CPT. RESP2 Redis: GET 195k rps, p50 < 0.3 ms, 42 tests. Fabric #2061 fixed a streaming-goroutine deadlock (Go, 42k stars).

SKILLS

Java & JVM	Java 17, Spring Boot/MVC, Guice, JUnit 5, Mockito, Maven · concurrency, GC profiling
Distributed	Microservices, REST, GraphQL, idempotency, state machines, DLQ, at-least-once, TDD
AWS	Lambda, DynamoDB, SQS, Step Functions, API Gateway, CloudWatch, ECS, CDK (Java)
Data	DynamoDB, Redis, NoSQL modelling, SQL, caching
Applied AI	Bedrock auth, streaming clients, Claude, Voxtral
Tooling	Git, CI/CD, JaCoCo gates, Agile, on-call, incident response · Python, C++

EXPERIENCE

Software Development Engineer, Jul 2022 — Sep 2025

Amazon · Travel (Hotels & Flights) · Bengaluru

- Concurrency. Parallelised 6 partner APIs via `CompletableFuture.allOf()` — p90 47% (1.1s→0.59s), conversion +33%; GZIP cut wire size 95%.
- Event-driven. Designed an idempotent flight schedule-change webhook (SQS, dedup keys) — 43 bps CPT; Step Functions booking state machine with Java Lambdas.
- Testing. Held 80%+ line coverage; `@SpringBootTest` + in-memory DynamoDB; introduced Gradle JaCoCo gates in CI.
- IaC. Authored service infra in **Java CDK**; migrated 2 services from CloudFormation — rollback incidents halved.
- Ownership. Owned 4 customer-facing microservices (search → order review); Spring Boot REST, DynamoDB DAOs, Prime and partner discount flows — ~30 bps funnel churn drop.

Software Development Engineer, Intern, May 2021 – Jul 2021 · Feb 2022 – Jul 2022

Amazon · Bengaluru

- Migrated GraphQL resolvers to Java + DynamoDB (API Gateway + Lambda); shipped **One-Click Renewal**; 90%+ unit-test coverage (JUnit 5 + Mockito).

PROJECTS

Redis Server in Java

github.com/praxstack/redis-server-java

RESP2 from scratch; bounded thread pool, `ConcurrentHashMap`, hybrid TTL. GET 195k rps, SET 154k rps, p50 < 0.3 ms; 42 tests.

Open Source — Fabric (42k stars)

danielmiessler/Fabric · Go

3 merged PRs: Bedrock bearer auth (#2044), dynamic regions (#2052), streaming deadlock fix (#2061).

Audio Transcription Pipeline

github.com/praxstack/audio-transcription-pipeline

Six-stage audio/video → speaker-labelled Markdown. Bedrock Voxtral ASR + Claude repair. 63 tests.

Warp BYOK Proxy

github.com/praxstack/warp-byok-proxy

Rust local proxy routing Warp Terminal AI calls to AWS Bedrock (BYOK, bearer and AWS credentials). 113 Rust tests. AGPL-3.0.

EDUCATION

Master of Computer Applications (M.C.A.), Computer Science · MNNIT, Prayagraj · GPA 9.25 · 2019 — 2022

Bachelor of Science (B.Sc. Hons), Computer Science · University of Delhi · 2015 — 2018

AWARDS

Amazon Pay Merchants Categories Champion — 100+ hotel trust-buster bugs; on-call 20+ to ~5 per shift · 2023